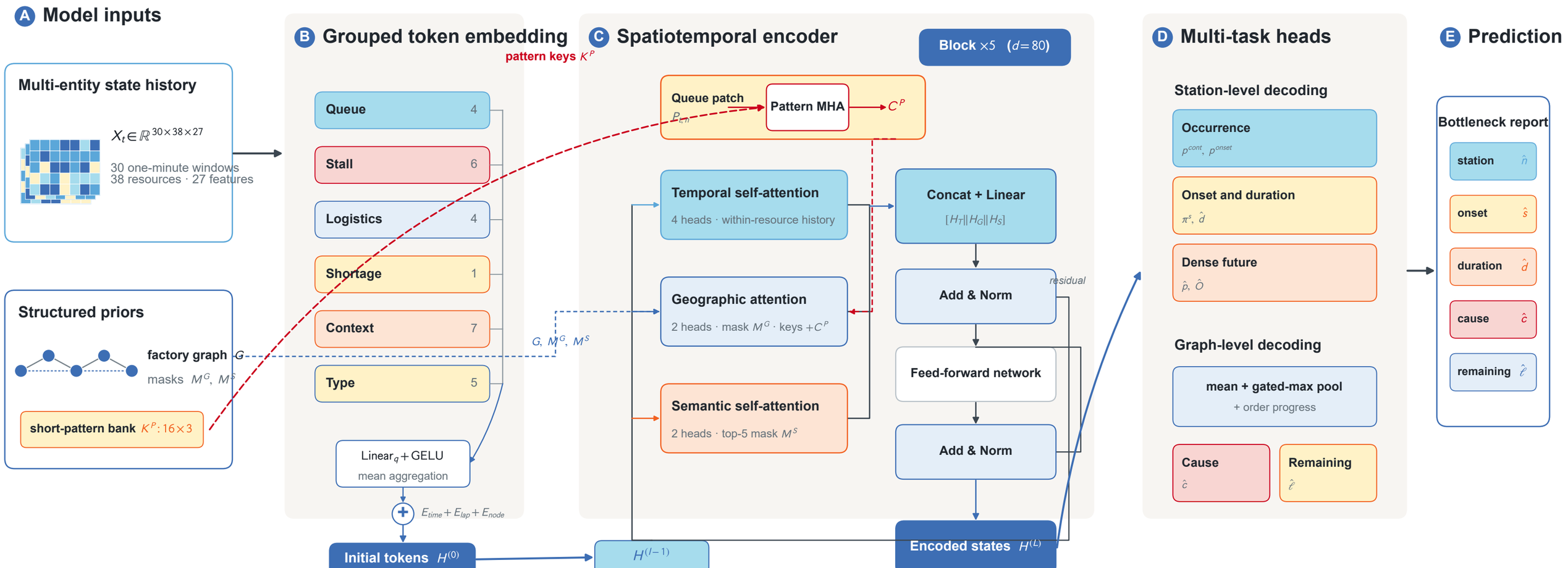

A Model inputs
Multi-entity state history
$X_t \in \mathbb{R}^{30\times38\times27}$
30 one-minute windows
38 resources · 27 features
Structured priors
factory graph G
masks M^G, M^S
short-pattern bank K^P: 16×3
B Grouped token embedding
pattern keys K^P
Queue 4
Stall 6
Logistics 4
Shortage 1
Context 7
Type 5
G, M^G, M^S
Linear$_q$ + GELU
mean aggregation
$E_{time} + E_{lap} + E_{node}$
Initial tokens $H^{(0)}$
C Spatiotemporal encoder
Block ×5 (d = 80)
Queue patch $P_{t,n}$
Pattern MHA
C^P
Temporal self-attention
4 heads · within-resource history
Geographic attention
2 heads · mask M^G · keys +C^P
Semantic self-attention
2 heads · top-5 mask M^S
Concat + Linear
$[H_T \| H_G \| H_S]$
Add & Norm
residual
Feed-forward network
Add & Norm
$H^{(l-1)}$
Encoded states $H^{(L)}$
D Multi-task heads
Station-level decoding
Occurrence
p^{cont}, p^{onset}
Onset and duration
π^s, $\hat{d}$
Dense future
$\hat{p}$, $\hat{O}$
Graph-level decoding
mean + gated-max pool
+ order progress
Cause
$\hat{c}$
Remaining
$\hat{\ell}$
E Prediction
Bottleneck report
station $\hat{n}$
onset $\hat{s}$
duration $\hat{d}$
cause $\hat{c}$
remaining $\hat{\ell}$